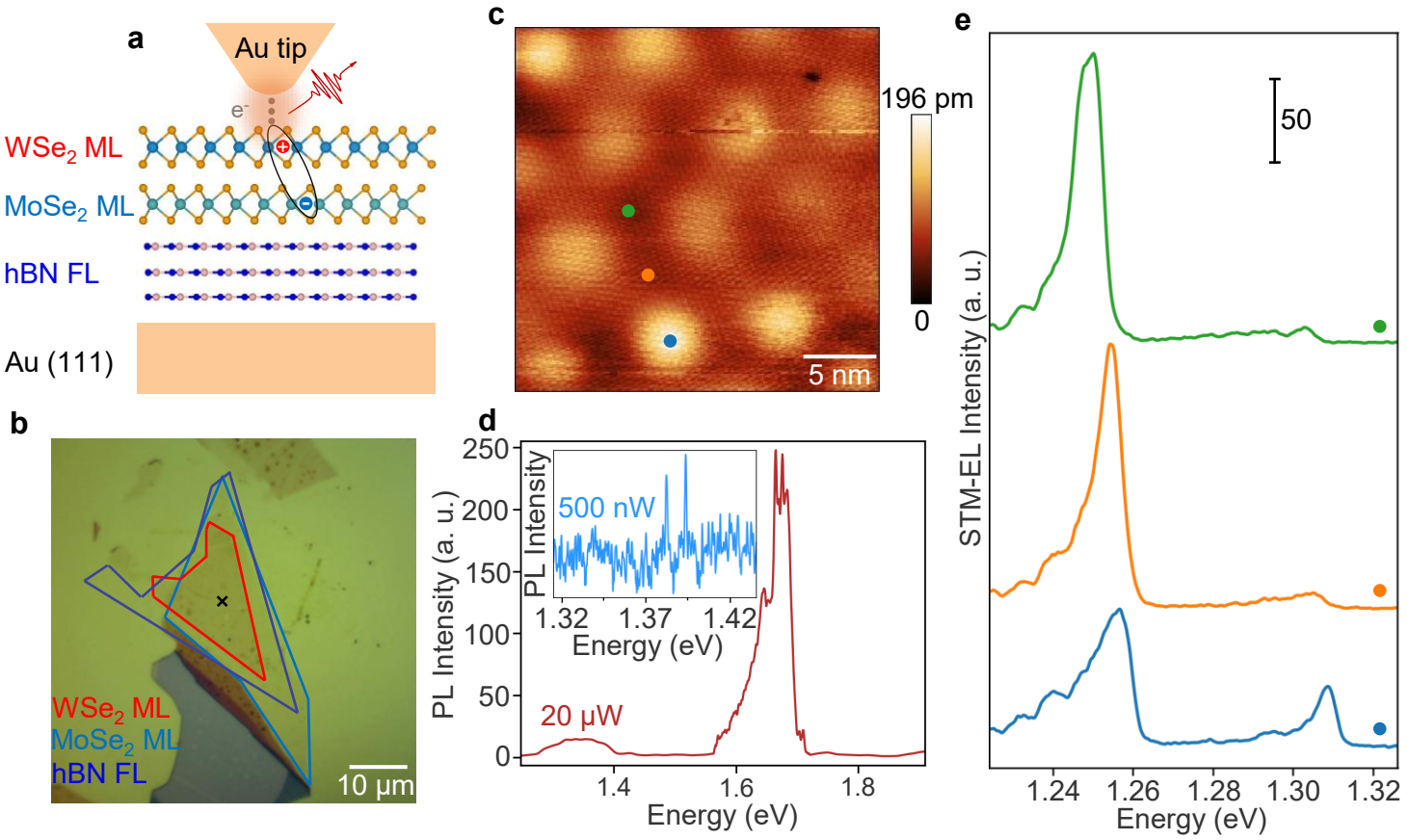


FIG. 1. STM electroluminescence of a $WSe_2/MoSe_2$/hBN heterostructure. a. Schematic of the scanning tunneling microscope electroluminescence (STM-EL) experiment. b. Optical microscopy image of the sample, showing the WSe_2 monolayer in red, the $MoSe_2$ monolayer in light blue, and the few-layer hBN in blue, with the underlying Au(111) substrate forming the background. c. Atomically resolved constant-current STM topography image of the heterostructure acquired at V = 1 V and I = 10 pA. d. Far field photoluminescence (PL) spectra acquired at 9 K under continuous-wave laser excitation at 633 nm from the cross-marked region in b. The inset shows high-resolution PL spectrum acquired from the same region at a lower excitation power of 500 nW. e. STM-EL spectrum acquired from the dotted region in c with V = −2.5 V, I = 100 pA.

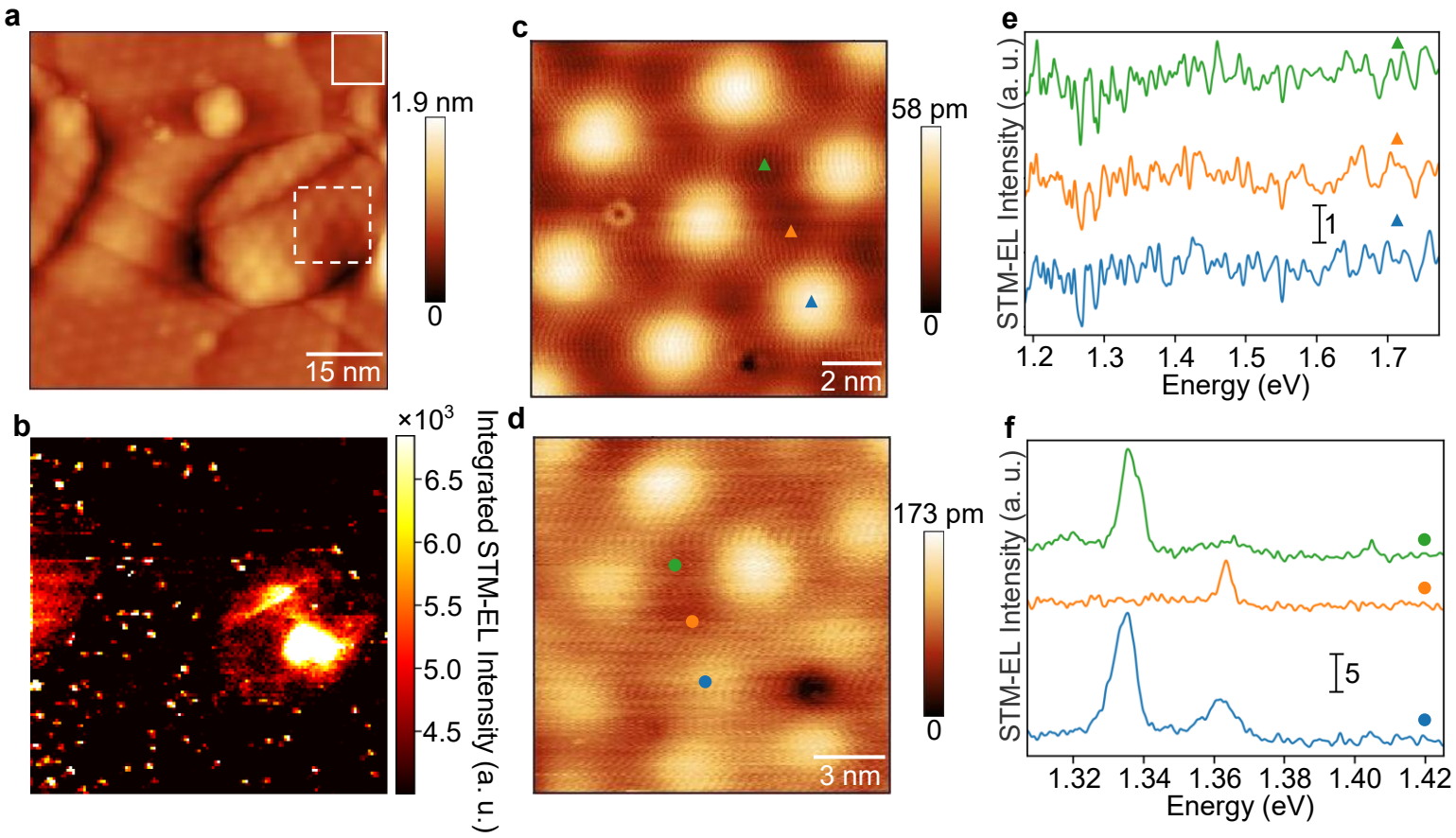


FIG. 2. Spatial mapping of STM electroluminescence in a $WSe_2/MoSe_2$ heterostructure. a. STM topography image of the the $WSe_2/MoSe_2$ surface acquired at V = 1 V and I = 10 pA. b. APD photon-intensity map of the same area acquired in constant-current mode (V = −2 V, I = 100 pA, with a pixel acquisition time of 300 ms). c. High-resolution constant-current STM topography images of the solid- and dashed-square regions, respectively, marked in a (V = 1 V and I = 10 pA) of the solid square box in a. d. High resolution constant-current STM image (V = 1 V and I = 10 pA) of the dashed square box in a. e. STM-EL spectrum acquired from the triangular-marked region in c, with V = −2.5 V, I = 1 nA. f. STM-EL spectrum acquired from the circled region in d, with V = −2.5 V, I = 100 pA.

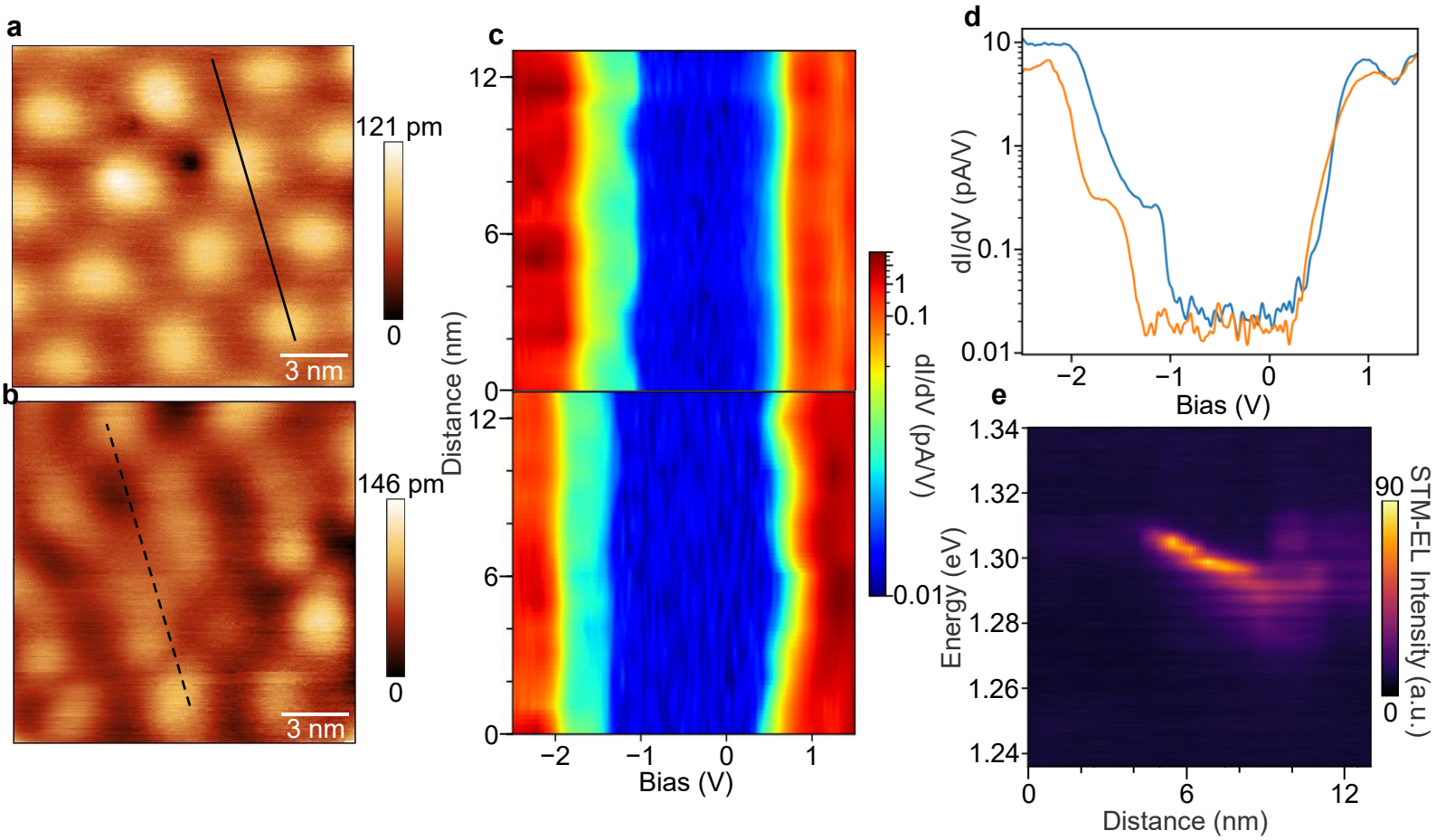


FIG. 3. Correlation between surface topography, electronic structure, and electroluminescence of a $WSe_2/MoSe_2$ heterostructure. a.,b. STM topography images of relatively flat and non-flat regions of the $WSe_2/MoSe_2$ surface, respectively, acquired at V = 1 V and I = 10 pA. c. STS line profiles (I = 100 pA) along the solid line in a and the dashed line in b, corresponding to the flat and non-flat regions, respectively. d. Representative differential conductance (dI/dV) spectra extracted from the STS line profiles in c for the flat (blue) and non-flat (orange) region. e. STM-EL spectrum recorded along the dashed line on a, with V = −2.5 V, I = 2 nA.

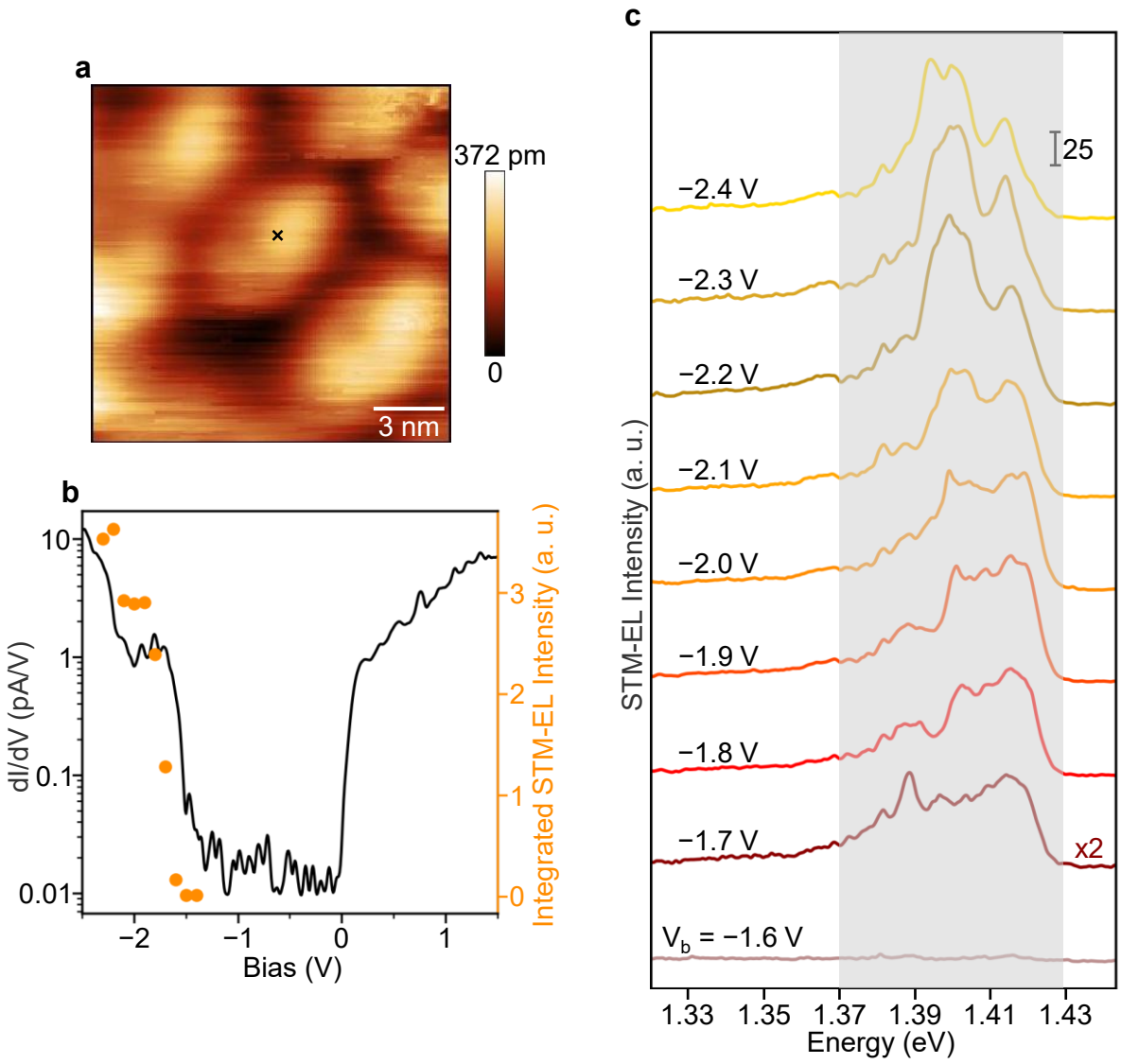


FIG. 4. Bias-dependent electronic spectroscopy and STM electroluminescence of the $WSe_2/MoSe_2$ heterostructure. a. STM topography image of the $WSe_2/MoSe_2$ surface (V = 1 V and I = 10 pA). b. Differential conductance (dI/dV) spectrum acquired from the cross-marked region in a, overlaid with the bias-dependent integrated STM-EL intensity of the spectral peak (shaded area) shown in c. c. STM-EL spectra recorded at different sample biases under constant-current conditions of I = 1 nA.